

Week 14: Bonus: type inference

Functional Programming in Lean

Learning goals

- ▶ Type constraints
- ▶ Constraint decomposition for simple types
- ▶ Occurs check

Core concepts

- ▶ Type variables and arrow types
- ▶ Arrow equality: input and output constraints
- ▶ Let-polymorphism: generalization and instantiation

Lean example

```
inductive Ty where  
  | nat  
  | variable (id : Nat)  
  | arrow (input output : Ty)
```

Lean details

- ▶ $\alpha \rightarrow \text{Nat} = \text{Nat} \rightarrow \beta$: two constraints
- ▶ $\alpha = \alpha \rightarrow \text{Nat}$: occurs-check failure
- ▶ Toy constraint solver versus Lean elaborator

Live coding

- ▶ Constraints for identity
- ▶ occurs
- ▶ Arrow-type decomposition

Check

Occurs-check result for $\alpha = \alpha \rightarrow \text{Nat}$

Exercise

- ▶ Constraint decomposition
- ▶ Let-polymorphism example